Theme and chassis layout

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Make a dimensioned layout that supports the circuit and drivetrain.

SUCCESS CRITERIA - I can...

1. Label component locations and clearances.
2. Use dimensions to check that parts fit.
3. Explain a structural choice with evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A 40-turn coil has flux per turn change by 0.006 Wb in 0.4 s. Find average induced-emf magnitude.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: available length = chassis length - components - clearances.
Use measured dimensions for a top view. Keep axles parallel and controls accessible. Get approval before tools/cuts.

Worked reasoning: A 20 cm chassis reserves 6 cm for the battery, 5 cm for the motor and 2 cm total for end clearances. The remaining 7 cm must cover other components and gaps.

Your turn: Draw a dimensioned top view showing these regions. Label any assumed dimensions rather than pretending they were measured.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Draw a dimensioned top view of your chassis. Label motor, battery, switch, axle lines and wheel clearance. Mark the intended forward direction.



Marzano check - circle: ① ② ③

04 Apply the physics

A sample chassis is 180 mm long. Reserve 15 mm at each end and a 95 mm component region. Mark the regions, calculate remaining length and convert the total chassis length to meters.



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Your theme adds a heavy decoration far to one side. Predict a possible effect and propose a way to test it without claiming the effect is certain.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 8 Ω resistor is in series with two parallel 12 Ω resistors across 28 V. Find parallel equivalent, total resistance and supply current.
Draw the stated circuit and label quantities relevant to the question.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Explain why a top-view sketch alone may miss a wheel or gear clearance problem. Add one useful side-view dimension.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 180 mm chassis needs two 15 mm clearances, a 60 mm battery and a 50 mm motor region. Find remaining length. Then identify a side-view clearance that this calculation misses.



Marzano check - circle: 1 2 3